

SMB 104: MICROBIAL BIOCHEMISTRY (New)

Purpose: To provide an overview of the nature and molecular structure of the macromolecules and their properties in biological systems.

Expected Learning Outcomes: By the end of the course, students will be able to:

- i) Describe the basic chemical and structural features of important classes of biomolecules, including proteins, lipids, carbohydrates and nucleic acids.
- ii) Differentiate the classes of carbohydrates, proteins, lipids and nucleic acids.
- iii) Identify the principles of molecules interactions and dynamics in microbial cellular systems.

Course Content:

Biomolecules and biopolymers: properties, classification and biological functions of the major and minor essential elements; water, carbohydrate, lipids, nucleic acids and proteins. Molecular architecture of microbial cells: chemical composition and function of cellular structures and organelles. Antimicrobial agents: type, mode of action and efficiency; antimicrobial resistance; effectiveness of antimicrobial agents; assay of antibiotics by chemical methods.

Contact hours: 35 hrs.

Mode of delivery

Lectures: tutorials, class discussions, student presentations, demonstrations and illustrations, written assignments

Practicals: course relevant experiments and field visits

Instructional Materials/Equipments: Course notes, class discussions and student presentations, demonstrations.

Course assessment:

Continuous assessment tests – 30 %; End of semester examination – 70%

Assessment of lecturer performance

Student-Lecturer evaluation e-forms

Pass Mark: 40 %

Course Reference Materials:**Core Textbooks**

- i) Prescott, L.M., Harley, J.P. and Klein, D.A. (2002). Microbiology. 5th Edition. Debuque Iowa, Wm C. Brown Publisher. ISBN: 9780697293909

- ii) Voet, D. J., Voet, G.J. and Pratt, C. W. (2008) Fundamentals of Biochemistry. 3rd Ed. J. Wiley and Sons Inc. ISBN: 9780470233962.
- iii) Mathews, C. K., Van Holde, K. E. and Ahern, K.G. (1999) Biochemistry. 3rd Ed. Prentice Hall. ISBN 10: 0805330666; ISBN 13: 9780805330663.

Recommended Textbooks

- i) Harper's Illustrated Biochemistry, 27th Ed. by R.K. Murray, D.K. Grannar, and V.W. Rodwell. McGraw Hill. ISBN-0-07-121766-5.
- ii) Lehninger, W. H. Nelson, D. L., Cox, M. M. (2012) Principles of Biochemistry 6th Ed. by Freeman Publishers ISBN-10: 1429234148; ISBN-13: 978-1429234146

Journals

- i) Biochemistry and Molecular Biology ISSN: 2471-8084
- ii) Annual Review of Biophysics and Biomolecular Structure ISSN:1936-122X
- iii) International Journal of Biochemistry and Molecular Biology ISSN 2152-4114

E-Materials

- i) <http://science.uokerbala.edu.iq/images/books/PLANT%20BIOCHEMISTRY%201.pdf>
- ii) <http://www.ncbi.nlm.nih.gov/books/NBK9879/>
- iii) http://www.cartercenter.org/resources/pdfs/health/ephti/library/lecture_notes/health_science_students/MedicalBiochemistry.pdf

Introduction

Biomolecules

Biomolecules are chemical compounds that occur naturally in living organisms. They are essential for maintaining life and performing biological processes such as metabolism, growth, and reproduction.

Key Features of Biomolecules

- ✓ Carbon-based compounds
- ✓ Exhibit structural diversity
- ✓ Participate in metabolic reactions
- ✓ Form complex macromolecules

Classification of Biomolecules

A. Water is the most abundant molecule in living systems.

Properties of Water

- Polar molecule
- High heat capacity
- Excellent solvent
- Cohesion and adhesion

Functions

- Medium for biochemical reactions
- Transport of substances
- Temperature regulation

Mineral Salts

Examples include Na^+ , K^+ , Ca^{2+} , Mg^{2+} , Cl^-

Functions

- Maintain osmotic balance
- Nerve impulse transmission
- Structural roles (bones, teeth)

ORGANIC BIOMOLECULES

Carbohydrates

Classification

1. Monosaccharides – glucose, fructose
2. Disaccharides – sucrose, lactose
3. Polysaccharides – starch, glycogen, cellulose

Properties

- Hydrophilic
- Easily metabolized

Functions

- Energy source
- Energy storage
- Structural support

Proteins

Structure Levels

- Primary
- Secondary
- Tertiary
- Quaternary

Properties

- Specificity
- Denaturation under extreme conditions

Functions

- Enzymatic catalysis
- Transport
- Immunity
- Structural support

Lipids

Types

- Triglycerides

- Phospholipids
- Steroids

Properties

- Hydrophobic
- High energy yield

Functions

- Energy storage
- Membrane structure
- Hormone synthesis

Nucleic Acids

Types

- DNA
- RNA

Functions

- Genetic information storage
- Protein synthesis

Biopolymers

Biopolymers are **large macromolecules formed by repeating monomers**.

Types of Biopolymers

Biopolymer	Monomer	Example	Function
Polysaccharides	Monosaccharides	Starch	Energy storage
Proteins	Amino acids	Enzymes	Catalysis
Nucleic acids	Nucleotides	DNA	Genetic control

Properties

- High molecular weight
- Specific 3D structure
- Functional specificity
- Can be linear or branched

Essential Elements in Biology

Living organisms require **chemical elements** for survival and metabolism.

Major (Primary) Elements

These make up ~96% of living matter:

- Carbon (C)
- Hydrogen (H)
- Oxygen (O)
- Nitrogen (N)

Properties

- Form stable covalent bonds
- Enable complex molecule formation

Functions

- Structural backbone of biomolecules
- Energy transformations (oxidation-reduction reactions)

B. Secondary Elements

- Calcium (Ca)
- Phosphorus (P)
- Potassium (K)
- Sulfur (S)
- Sodium (Na)
- Chlorine (Cl)
- Magnesium (Mg)

Functions

- Bone formation (Ca, P)
- Nerve impulse transmission (Na, K)
- Enzyme cofactors (Mg)
- Protein structure (S in amino acids)

C. Trace (Minor) Elements

Required in very small amounts:

- Iron (Fe)
- Zinc (Zn)

- Copper (Cu)
- Iodine (I)
- Selenium (Se)
- Manganese (Mn)

Functions

- Enzyme activation (Zn, Mn)
- Oxygen transport (Fe in hemoglobin)
- Hormone synthesis (I in thyroid hormones)

Properties of Biomolecules (General)

Chemical Properties

- Reactivity depends on functional groups ($-\text{OH}$, $-\text{COOH}$, $-\text{NH}_2$)
- Participate in metabolic pathways

Physical Properties

- Solubility varies (polar vs non-polar)
- Stability influenced by temperature and pH

Biological Properties

- Specificity (enzymes)
- Self-assembly (membranes)

Regulation and signaling INTERACTIONS OF BIOMOLECULES

- Hydrogen bonding
- Ionic interactions
- Van der Waals forces
- Hydrophobic interactions

These interactions determine structure and function.

METABOLIC ROLES

Biomolecules are involved in:

- Catabolism (breakdown)
- Anabolism (synthesis)

Energy is transferred via ATP.

IMPORTANCE IN MICROBIAL BIOCHEMISTRY

- Microorganisms depend on biomolecules for survival
- Enzymes regulate metabolic pathways
- Nutrient cycling in ecosystems

SUMMARY

Biomolecules form the basis of life. Their classification into carbohydrates, proteins, lipids, and nucleic acids helps in understanding their diverse roles. Essential elements support structure and metabolism. Biopolymers provide structural and functional integrity to living systems.

Elements in Living Cells

The most abundant element in cells is hydrogen (H), followed by carbon (C), oxygen (O), nitrogen (N), phosphorous (P), and sulfur (S). These are elements called **macronutrients**, and they account for about 99% of the dry weight of cells. Some elements, such as sodium (Na), potassium (K), magnesium (Mg), zinc (Zn), iron (Fe), calcium (Ca), molybdenum (Mo), copper (Cu), cobalt (Co), manganese (Mn), or vanadium (Va), are required by some cells in very small amounts and are called **micronutrients** or **trace elements**. All of these elements are essential to the function of many biochemical reactions, and, therefore, are essential to life.

The four most abundant elements in living matter (C, N, O, and H) have low atomic numbers and are thus light elements capable of forming strong bonds with other atoms to produce molecules.

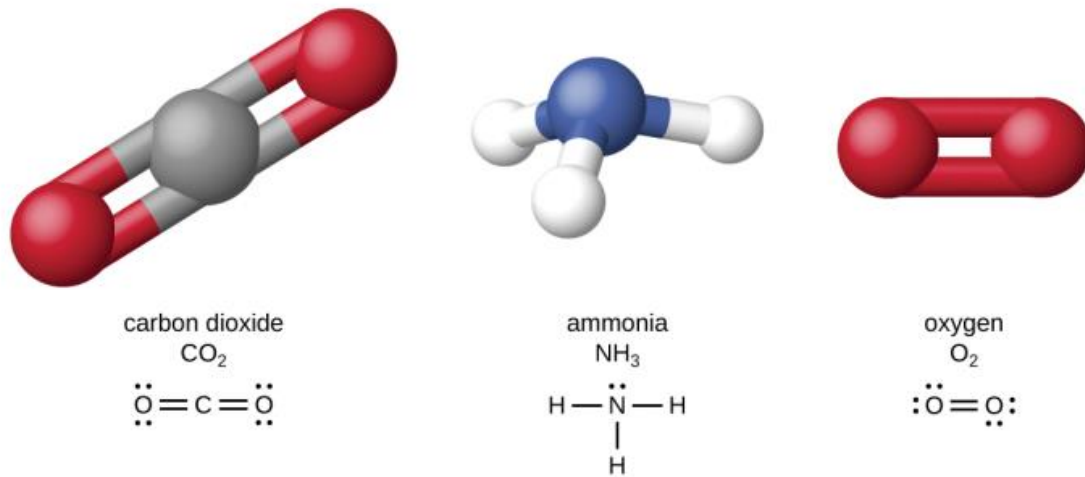


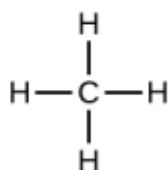
figure 7.2 Some common molecules include carbon dioxide, ammonia, and oxygen, which consist of combinations of oxygen atoms (red spheres), carbon atoms (gray spheres), hydrogen atoms (white spheres), or nitrogen atoms (blue spheres).

- ✓ Carbon forms four chemical bonds, whereas nitrogen forms three, oxygen forms two, and hydrogen forms one. When bonded together within molecules, oxygen, sulfur, and nitrogen often have one or more “lone pairs” of electrons that play important roles in determining many of the molecules’ physical and chemical properties. These traits in combination permit the formation of a vast number of diverse molecular species necessary to form the structures and enable the functions of living organisms.
- ✓ Living organisms contain inorganic compounds (mainly water and salts; and organic molecules. Organic molecules contain carbon; inorganic compounds do not. Carbon oxides and carbonates are exceptions; they contain carbon but are considered inorganic because they do not contain hydrogen.
- ✓ The atoms of an organic molecule are typically organized around chains of carbon atoms.
- ✓ Inorganic compounds make up 1%–1.5% of a living cell’s mass. They are small, simple compounds that play important roles in the cell, although they do not form cell structures. Most of the carbon found in organic molecules originates from inorganic carbon sources such as carbon dioxide captured via carbon fixation by microorganisms.

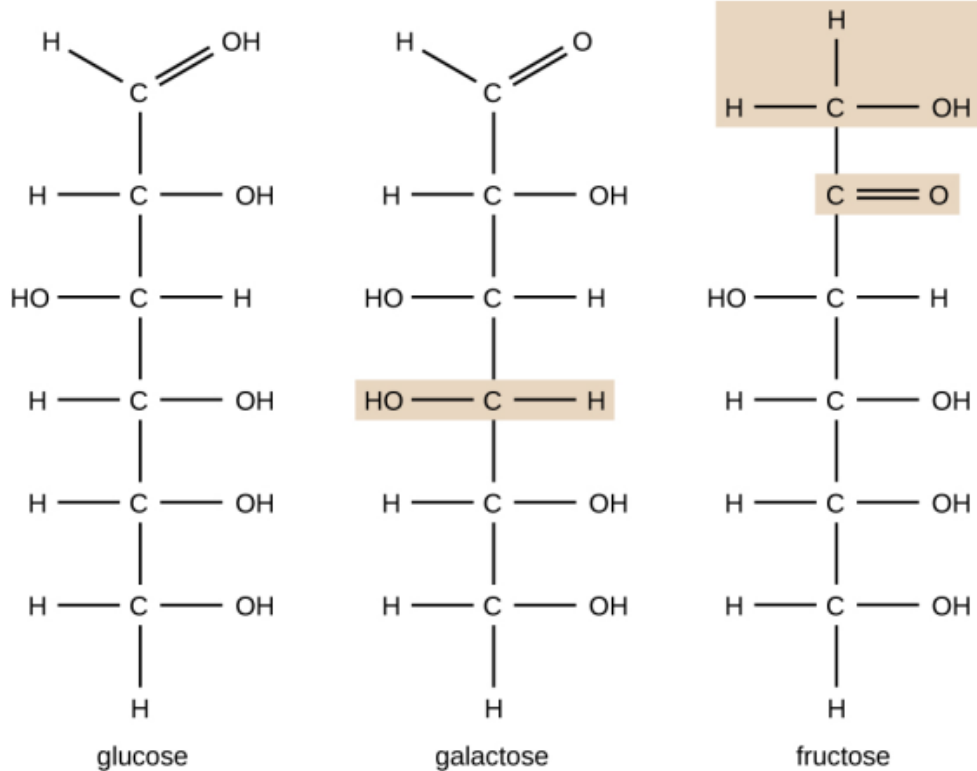
Organic Molecules and Isomerism

- ✓ Organic molecules in organisms are generally larger and more complex than inorganic molecules. Their carbon skeletons are held together by covalent bonds. They form the cells of an organism and perform the chemical reactions that facilitate life. All of these molecules, called **biomolecules** because they are part of living matter, contain carbon, which is the building block of life. Carbon is a very unique element in that it has four valence electrons in its outer orbitals and can form four single covalent bonds with up to four other

atoms at the same time. These atoms are usually oxygen, hydrogen, nitrogen, sulfur, phosphorous, and carbon itself; the simplest organic compound is methane, in which carbon binds only to hydrogen'



- ✓
- ✓ As a result of carbon's unique combination of size and bonding properties, carbon atoms can bind together in large numbers, thus producing a chain or **carbon skeleton**. The carbon skeleton of organic molecules can be straight, branched, or ring shaped (cyclic).
- ✓ Organic molecules are built on chains of carbon atoms of varying lengths; most are typically very long, which allows for a huge number and variety of compounds. No other element has the ability to form so many different molecules of so many different sizes and shapes.
- ✓ Molecules with the same atomic makeup but different structural arrangement of atoms are called **isomers**.
- ✓ The concept of isomerism is very important in chemistry because the structure of a molecule is always directly related to its function. Slight changes in the structural arrangements of atoms in a molecule may lead to very different properties.
- ✓ Chemists represent molecules by their **structural formula**, which is a graphic representation of the molecular structure, showing how the atoms are arranged.
- ✓ Compounds that have identical molecular formulas but differ in the bonding sequence of the atoms are called **structural isomers**. The monosaccharides glucose, galactose, and fructose all have the same molecular formula, C₆H₁₂O₆, but we can see from the figure that the atoms are bonded together differently.
- ✓ Glucose, galactose, and fructose have the same chemical formula (C₆H₁₂O₆), but these structural isomers differ in their physical and chemical properties.



Biologically Significant Functional Groups

- ✓ In addition to containing carbon atoms, biomolecules also contain **functional groups**—groups of atoms within molecules that are categorized by their specific chemical composition and the chemical reactions they perform, regardless of the molecule in which the group is found. Some of the most common functional groups are listed in **the table below**

Common Functional Groups Found in Biomolecules		
Name	Functional Group	Compounds
Aldehyde	$\text{R}-\overset{\text{O}}{\parallel}\text{C}-\text{H}$	Carbohydrates
Amide	$\text{R}-\overset{\text{O}}{\parallel}\text{C}-\text{N}-\text{R}'$ H	Proteins
Amino	$\text{R}-\text{NH}_2$	Amino acids, proteins
Carbonyl	$\text{R}-\overset{\text{O}}{\parallel}\text{C}-\text{R}'$	Ketones, aldehydes, carboxylic acids, amides
Carboxylic acid	$\text{R}-\overset{\text{O}}{\parallel}\text{C}-\text{O}-\text{H}$	Amino acids, proteins, fatty acids
Ester	$\text{R}-\overset{\text{O}}{\parallel}\text{C}-\text{O}-\text{R}'$	Lipids, nucleic acids
Ether	$\text{R}-\text{O}-\text{R}'$	Disaccharides, polysaccharides, lipids
Hydroxyl	$\text{R}-\text{O}-\text{H}$	Alcohols, monosaccharides, amino acids, nucleic acids
Ketone	$\text{R}-\overset{\text{O}}{\parallel}\text{C}-\text{R}'$	Carbohydrates
Methyl	$\text{R}-\text{CH}_3$	Methylated compounds such as methyl alcohols and methyl esters
Phosphate	$\text{R}-\text{PO}_3\text{H}_2$	Nucleic acids, phospholipids, ATP
Sulfhydryl	$\text{R}-\text{S}-\text{H}$	Amino acids, proteins

✓

- ✓ In the formulas, the symbol R stands for “residue” and represents the remainder of the molecule. R might symbolize just a single hydrogen atom or it may represent a group of many

atoms. Notice that some functional groups are relatively simple, consisting of just one or two atoms, while some comprise two of these simpler functional groups. For example, a carbonyl group is a functional group composed of a carbon atom double bonded to an oxygen atom: $C=O$. It is present in several classes of organic compounds as part of larger functional groups such as ketones, aldehydes, carboxylic acids, and amides. In ketones, the carbonyl is present as an internal group, whereas in aldehydes it is a terminal group.

Macromolecules

Carbon chains form the skeletons of most organic molecules. Functional groups combine with the chain to form biomolecules. Because these biomolecules are typically large, we call them **macromolecules**. Many biologically relevant macromolecules are formed by linking together a great number of identical, or very similar, smaller organic molecules. The smaller molecules act as building blocks and are called **monomers**, and the macromolecules that result from their linkage are called **polymers**. Cells and cell structures include four main groups of carbon-containing macromolecules: polysaccharides, proteins, lipids, and nucleic acids

- ✓ Of the many possible ways that monomers may be combined to yield polymers, one common approach encountered in the formation of biological macromolecules is **dehydration synthesis**. In this chemical reaction, monomer molecules bind end to end in a process that results in the formation of water molecules as a byproduct:

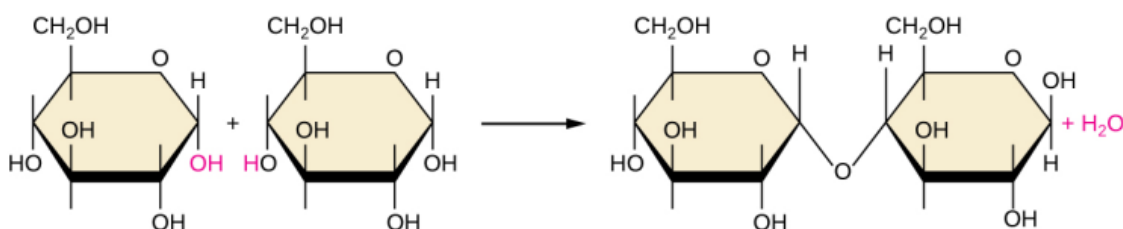
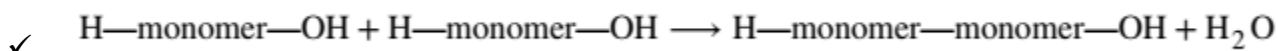


Figure 7.7 In this dehydration synthesis reaction, two molecules of glucose are linked together to form maltose. In the process, a water molecule is formed.

Some Functions of Macromolecules

Macromolecule	Functions
Carbohydrates	Energy storage, receptors, food, structural role in plants, fungal cell walls, exoskeletons of insects
Lipids	Energy storage, membrane structure, insulation, hormones, pigments
Nucleic acids	Storage and transfer of genetic information
Proteins	Enzymes, structure, receptors, transport, structural role in the cytoskeleton of a cell and the extracellular matrix

CARBOHYDRATES

The most abundant biomolecules on earth are **carbohydrates**. From a chemical viewpoint, carbohydrates are primarily a combination of carbon and water, and many of them have the empirical formula $(CH_2O)_n$, where n is the number of repeated units. This view represents these molecules simply as “hydrated” carbon atom chains in which water molecules attach to each carbon atom, leading to the term “carbohydrates.” Although all carbohydrates contain carbon, hydrogen, and oxygen, there are some that also contain nitrogen, phosphorus, and/or sulfur. Carbohydrates have myriad different functions.

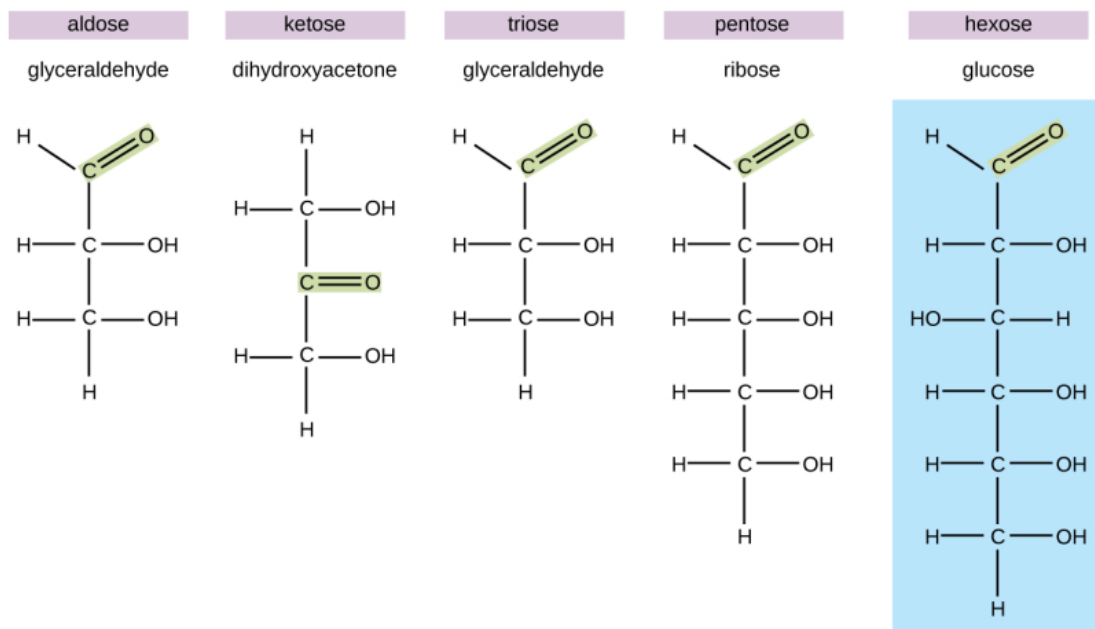
- They are abundant in terrestrial ecosystems, many forms of which we use as food sources.
- These molecules are also vital parts of macromolecular structures that store and transmit genetic information (i.e., DNA and RNA).
- They are the basis of biological polymers that impart strength to various structural components of organisms (e.g., cellulose and chitin),
- and they are the primary source of energy storage in the form of starch and glycogen.

Monosaccharides: The Sweet Ones

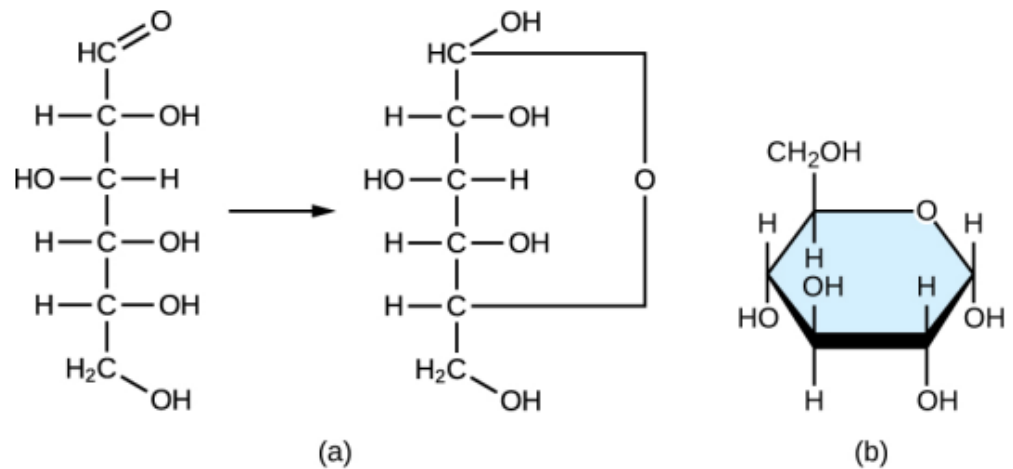
- ✓ In biochemistry, carbohydrates are often called **saccharides**, from the Greek **sakcharon**, meaning sugar, although not all the saccharides are sweet.

- ✓ The simplest carbohydrates are called **monosaccharides**, or simple sugars. They are the building blocks (monomers) for the synthesis of polymers or complex carbohydrates,
- ✓ Monosaccharides are classified based on the number of carbons in the molecule. General categories are identified using a prefix that indicates the number of carbons and the suffix – **ose**, which indicates a saccharide; for example, triose (three carbons), tetrose (four carbons), pentose (five carbons), and hexose (six carbons). The hexose D-glucose is the most abundant monosaccharide in nature. Other very common and abundant hexose monosaccharides are galactose, used to make the disaccharide milk sugar lactose, and the fruit sugar fructose.

Monosaccharides

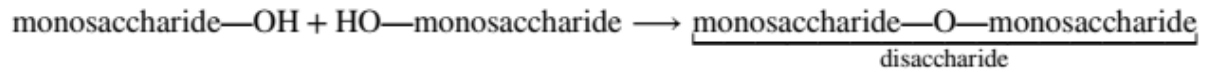


- ✓ Monosaccharides of four or more carbon atoms are typically more stable when they adopt cyclic, or ring, structures. These ring structures result from a chemical reaction between functional groups on opposite ends of the sugar's flexible carbon chain, namely the carbonyl group and a relatively distant hydroxyl group. Glucose, for example, forms a six-membered ring.



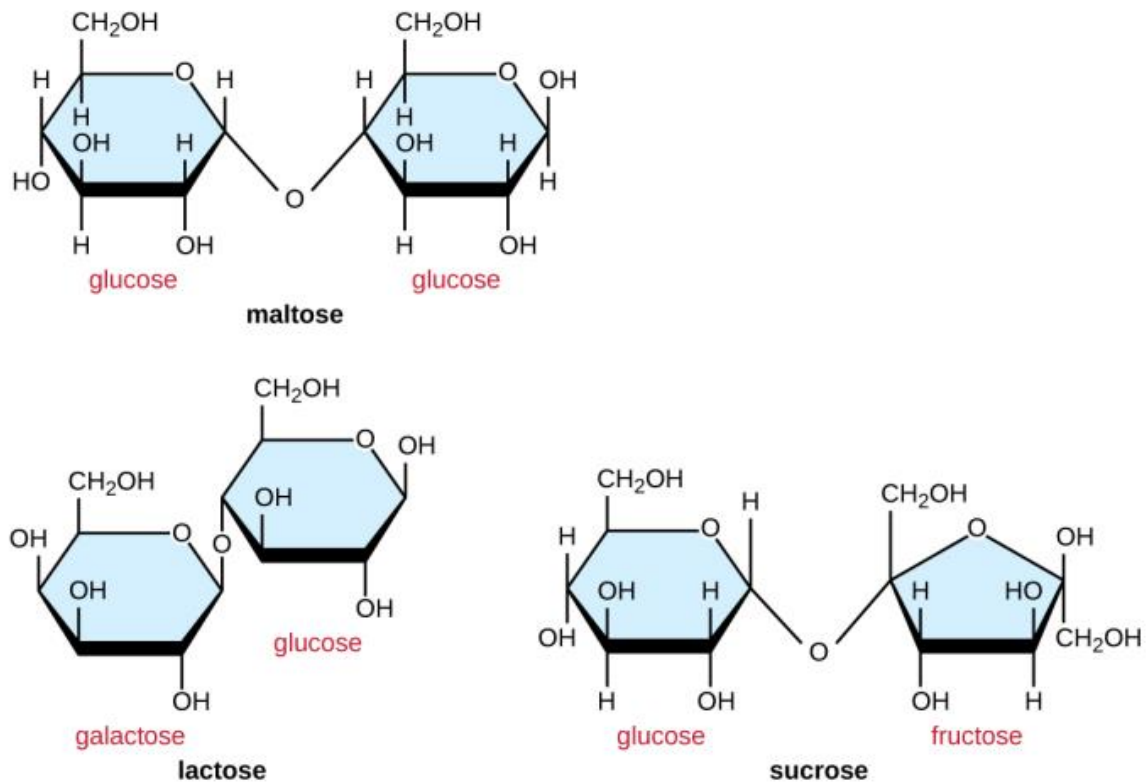
Disaccharides

- ✓ Two monosaccharide molecules may chemically bond to form a **disaccharide**. The name given to the covalent bond between the two monosaccharides is a **glycosidic bond**. Glycosidic bonds form between hydroxyl groups of the two saccharide molecules, an example of the dehydration synthesis



- ✓ Common disaccharides are the grain sugar maltose, made of two glucose molecules; the milk sugar lactose, made of a galactose and a glucose molecule; and the table sugar sucrose, made of a glucose and a fructose molecule.

✓

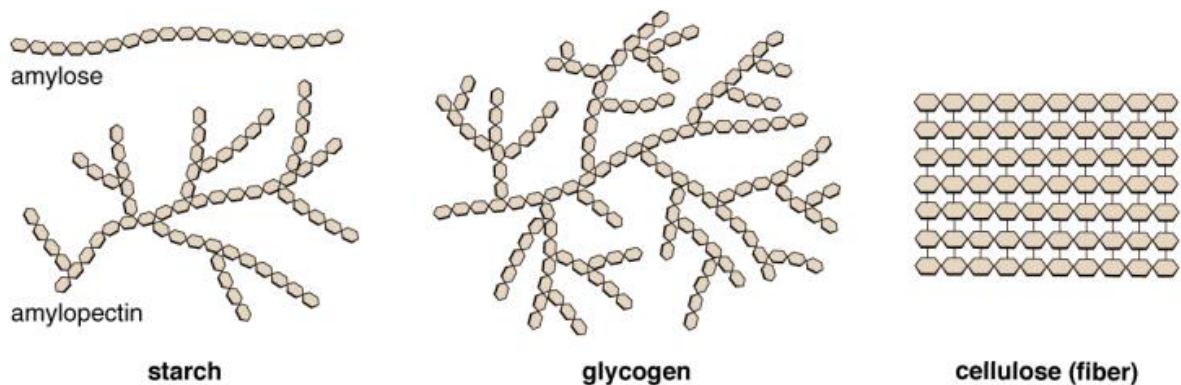


Polysaccharides

- ✓ Polysaccharides, also called glycans, are large polymers composed of hundreds of monosaccharide monomers. Unlike mono- and disaccharides, **polysaccharides** are not sweet and, in general, they are not soluble in water. Like disaccharides, the monomeric units of polysaccharides are linked together by glycosidic bonds.
- ✓ Polysaccharides are very diverse in their structure. Three of the most biologically important polysaccharides—**starch**, **glycogen**, and **cellulose**—are all composed of repetitive glucose units, although they differ in their structure (**Figure 7.11**). Cellulose consists of a linear chain of glucose molecules and is a common structural component of cell walls in plants and other organisms. Glycogen and starch are branched polymers; glycogen is the primary energy-storage molecule in animals and bacteria, whereas

plants primarily store energy in starch. The orientation of the glycosidic linkages in these three polymers is different as well and, as a consequence, linear and branched macromolecules have different properties.

- ✓ Modified glucose molecules can be fundamental components of other structural polysaccharides. Examples of these types of structural polysaccharides are N-acetyl glucosamine (NAG) and N-acetyl muramic acid (NAM) found in bacterial cell wall peptidoglycan. Polymers of NAG form chitin, which is found in fungal cell walls and in the exoskeleton of insects.



✓

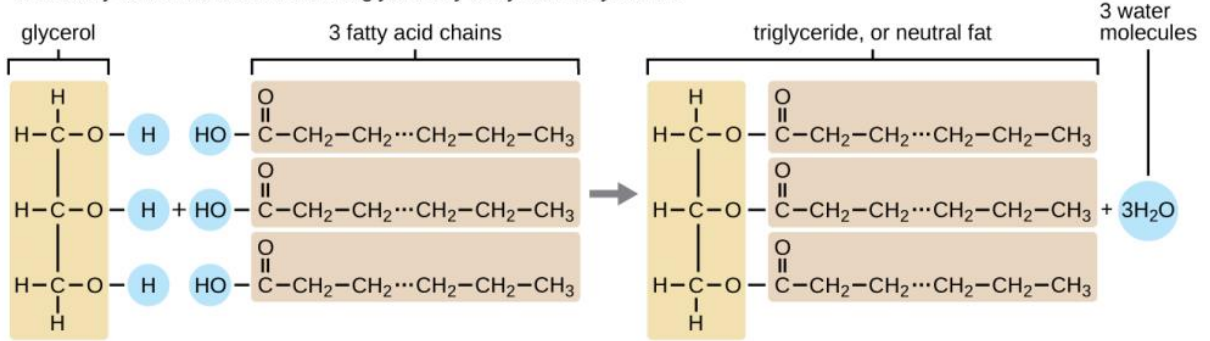
LIPIDS

Lipids are composed primarily of carbon and hydrogen, they may also contain oxygen, nitrogen, sulfur, and phosphorous. Lipids serve numerous and diverse purposes in the structure and functions of organisms. They can be a source of nutrients, a storage form for carbon, energy-storage molecules, or structural components of membranes and hormones. Lipids comprise a broad class of many chemically distinct compounds as shown below.

Fatty Acids and Triacylglycerides

- ✓ The **fatty acids** are lipids that contain long-chain hydrocarbons terminated with a carboxylic acid functional group. Because of the long hydrocarbon chain, fatty acids are **hydrophobic** (“water fearing”) or nonpolar.
- ✓ Fatty acids with hydrocarbon chains that contain only single bonds are called **saturated fatty acids** because they have the greatest number of hydrogen atoms possible and are, therefore, “saturated” with hydrogen.
- ✓ Fatty acids with hydrocarbon chains containing at least one double bond are called **unsaturated fatty acids** because they have fewer hydrogen atoms.
- ✓ Saturated fatty acids have a straight, flexible carbon backbone, whereas unsaturated fatty acids have “kinks” in their carbon skeleton because each double bond causes a rigid bend of the carbon skeleton. These differences in saturated versus unsaturated fatty acid structure result in different properties for the corresponding lipids in which the fatty acids are incorporated.
- ✓ For example, lipids containing saturated fatty acids are solids at room temperature, whereas lipids containing unsaturated fatty acids are liquids.
- ✓ A **triacylglycerol**, or **triglyceride**, is formed when three fatty acids are chemically linked to a glycerol molecule. Triglycerides are the primary components of adipose tissue (body fat), and are major constituents of sebum (skin oils). They play an important metabolic role, serving as efficient energy-storage molecules that can provide more than double the caloric content of both carbohydrates and proteins.

Three fatty acid chains are bound to glycerol by dehydration synthesis.



Phospholipids and Biological Membranes

- ✓ Triglycerides are classified as simple lipids because they are formed from just two types of compounds: glycerol and fatty acids.
- ✓ In contrast, complex lipids contain at least one additional component, for example, a phosphate group (**phospholipids**) or a carbohydrate moiety (**glycolipids**).
- ✓ A typical phospholipid composed of two fatty acids linked to glycerol (a diglyceride). The two fatty acid carbon chains may be both saturated, both unsaturated, or one of each. Instead of another fatty acid molecule (as for triglycerides), the third binding position on the glycerol molecule is occupied by a modified phosphate group.
- ✓ The molecular structure of lipids results in unique behavior in aqueous environments.

Figure 7.12 depicts the structure of a triglyceride. Because all three substituents on the glycerol backbone are long hydrocarbon chains, these compounds are nonpolar and not significantly attracted to polar water molecules—they are hydrophobic. Conversely, phospholipids such as the one shown in **Figure 7.13** have a negatively charged phosphate group. Because the phosphate is charged, it is capable of strong attraction to water molecules and thus is hydrophilic, or “water loving.” The hydrophilic portion of the

phospholipid is often referred to as a polar “head,” and the long hydrocarbon chains as nonpolar “tails.” A molecule presenting a hydrophobic portion and a hydrophilic moiety is said to be amphipathic. Notice the “R” designation within the hydrophilic head depicted in **Figure 7.13**, indicating that a polar head group can be more complex than a simple phosphate moiety. Glycolipids are examples in which carbohydrates are bonded to the lipids’ head groups.

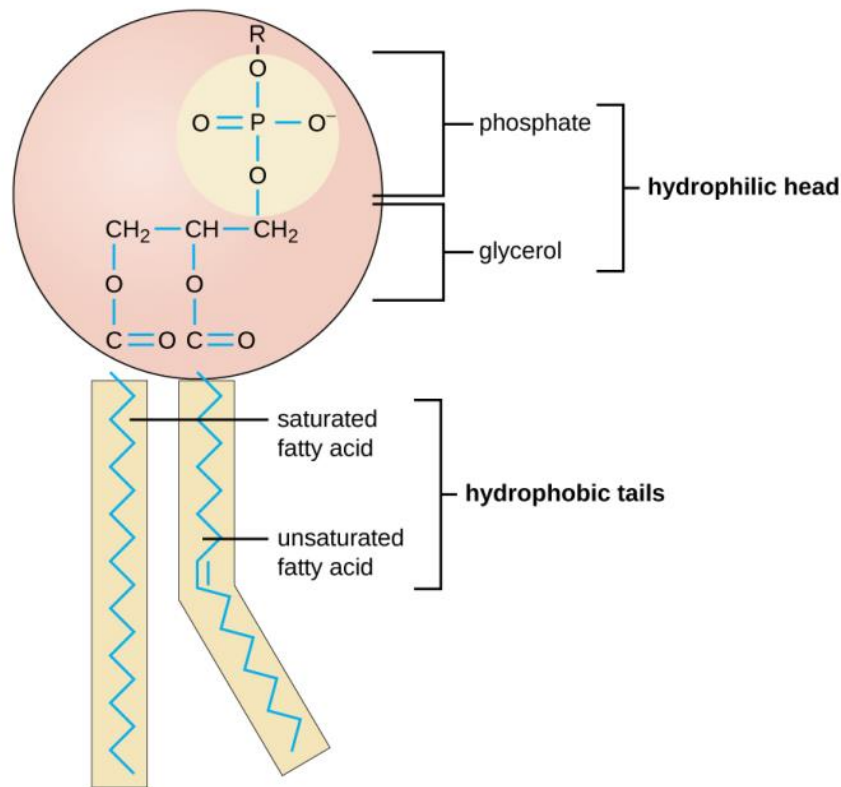


Figure 7.13 This illustration shows a phospholipid with two different fatty acids, one saturated and one unsaturated, bonded to the glycerol molecule. The unsaturated fatty acid has a slight kink in its structure due to the double bond.

- ✓ The amphipathic nature of phospholipids enables them to form uniquely functional structures in aqueous environments. the polar heads of these molecules are strongly attracted to water molecules, and the nonpolar tails are not.

- ✓ Because of their considerable lengths, these tails are, in fact, strongly attracted to one another. As a result, energetically stable, large-scale assemblies of phospholipid molecules are formed in which the hydrophobic tails congregate within enclosed regions, shielded from contact with water by the polar heads (Figure 7.14).

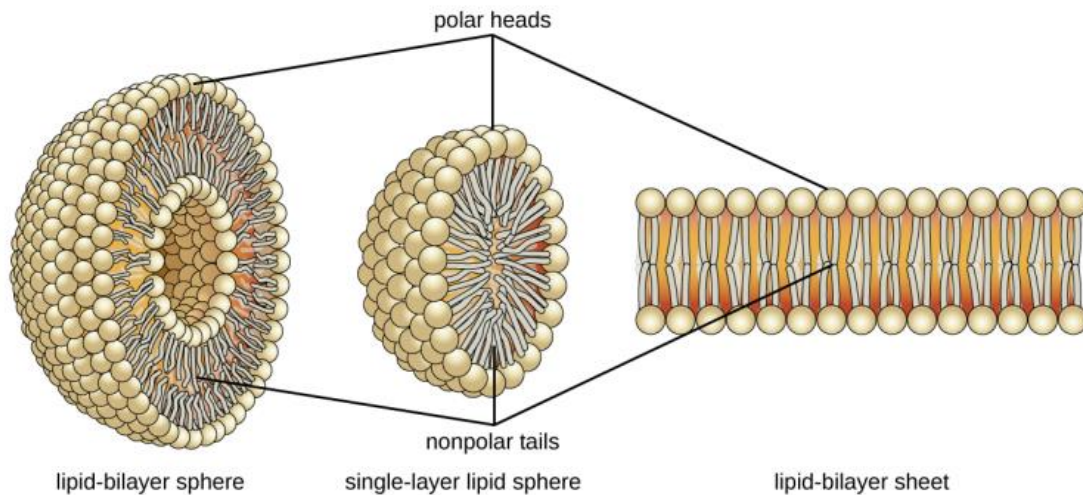


Figure 7.14 Phospholipids tend to arrange themselves in aqueous solution forming liposomes, micelles, or lipid bilayer sheets. (credit: modification of work by Mariana Ruiz Villarreal)

- ✓ The simplest of these structures are micelles, spherical assemblies containing a hydrophobic interior of phospholipid tails and an outer surface of polar head groups. Larger and more complex structures are created from lipid-bilayer sheets, or unit membranes, which are large, two-dimensional assemblies of phospholipids congregated tail to tail.
- ✓ The cell membranes of nearly all organisms are made from lipid-bilayer sheets, as are the membranes of many intracellular components. These sheets may also form lipid-bilayer spheres that are the structural basis of vesicles and liposomes, subcellular components that play a role in numerous physiological functions.

Isoprenoids and Sterols

- ✓ The **isoprenoids** are branched lipids, also referred to as terpenoids, that are formed by chemical modifications of the isoprene molecule
- ✓ These lipids play a wide variety of physiological roles in plants and animals, with many technological uses as pharmaceuticals (capsaicin), pigments (e.g., orange beta carotene, xanthophylls), and fragrances (e.g., menthol, camphor, limonene [lemon fragrance], and pinene [pine fragrance]).
- ✓ Long-chain isoprenoids are also found in hydrophobic oils and waxes. Waxes are typically water resistant and hard at room temperature, but they soften when heated and liquefy if warmed adequately.
- ✓ In humans, the main wax production occurs within the sebaceous glands of hair follicles in the skin, resulting in a secreted material called sebum, which consists mainly of triacylglycerol, wax esters, and the hydrocarbon squalene. There are many bacteria in the microbiota on the skin that feed on these lipids. One of the most prominent bacteria that feed on lipids is **Propionibacterium acnes**, which uses the skin's lipids to generate short-chain fatty acids and is involved in the production of acne.

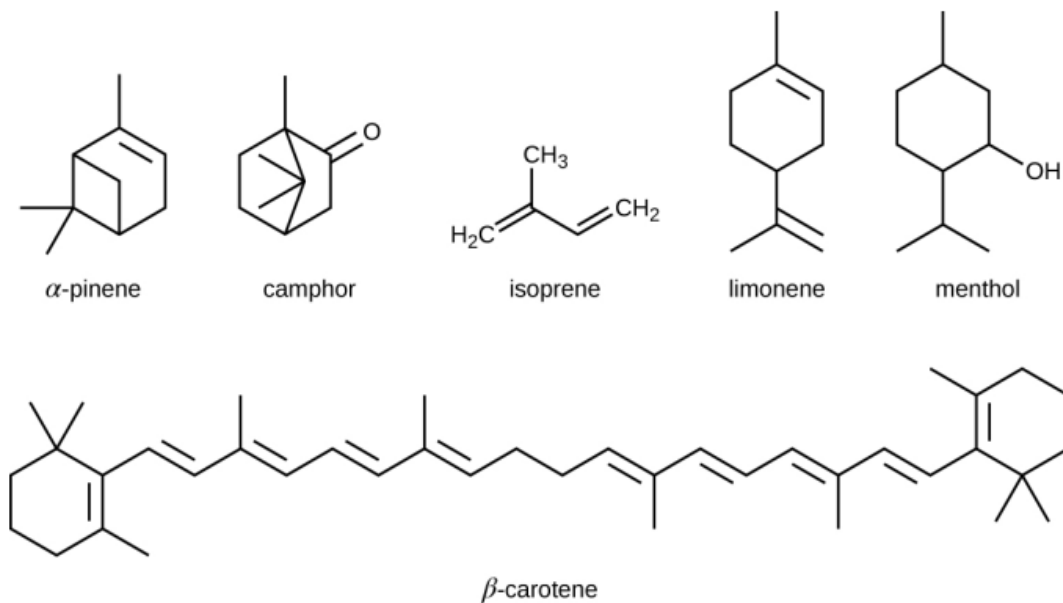


Figure 7.15 Five-carbon isoprene molecules are chemically modified in various ways to yield isoprenoids.

Steroids

- ✓ These are complex, ringed structures that are found in cell membranes; some function as hormones. The most common types of steroids are **sterols**, which are steroids containing an OH group. These are mainly hydrophobic molecules, but also have hydrophilic hydroxyl groups.
- ✓ The most common sterol found in animal tissues is cholesterol. Its structure consists of four rings with a double bond in one of the rings, and a hydroxyl group at the sterol-defining position. The function of cholesterol is to strengthen cell membranes in eukaryotes and in bacteria without cell walls, such as *Mycoplasma*.
- ✓ Prokaryotes generally do not produce cholesterol, although bacteria produce similar compounds called hopanoids, which are also multiringed structures that strengthen bacterial membranes (Figure 7.16). Fungi and some protozoa produce a similar compound called ergosterol, which strengthens the cell membranes of these organisms.

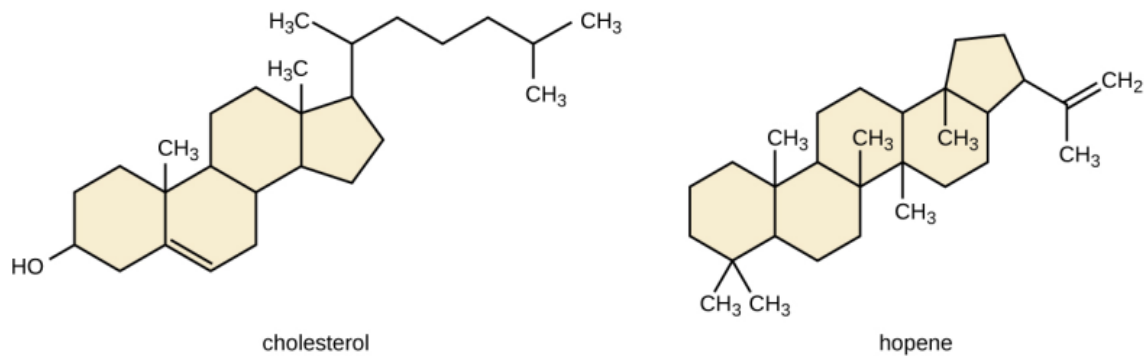
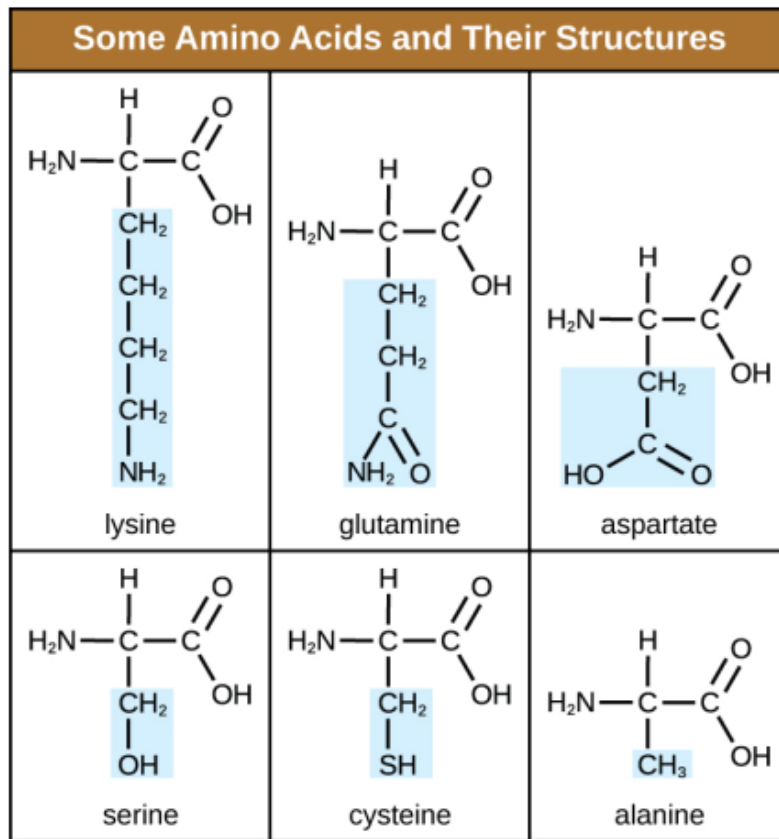


Figure 7.16 Cholesterol and hopene (a hopanoid compound) are molecules that reinforce the structure of the cell membranes in eukaryotes and prokaryotes, respectively.

PROTEINS

Amino Acids and Peptide Bonds

- ✓ An **amino acid** is an organic molecule in which a hydrogen atom, a carboxyl group (COOH), and an amino group (NH_2) are all bonded to the same carbon atom, the so-called α carbon.
- ✓ The fourth group bonded to the α carbon varies among the different amino acids and is called a residue or a **side chain**, represented in structural formulas by the letter **R**.
- ✓ A residue is a monomer that results when two or more amino acids combine and remove water molecules.
- ✓ The primary structure of a protein, a peptide chain, is made of amino acid residues.
- ✓ The unique characteristics of the functional groups and **R** groups allow these components of the amino acids to form hydrogen, ionic, and disulfide bonds, along with polar/nonpolar interactions needed to form secondary, tertiary, and quaternary protein structures.
- ✓ These groups are composed primarily of carbon, hydrogen, oxygen, nitrogen, and sulfur, in the form of hydrocarbons, acids, amides, alcohols, and amine



*Blue shading indicates R group.

- ✓ Amino acids may chemically bond together by reaction of the carboxylic acid group of one molecule with the amine group of another. This reaction forms a **peptide bond** and a water molecule and is another example of dehydration synthesis (Figure 7.18).

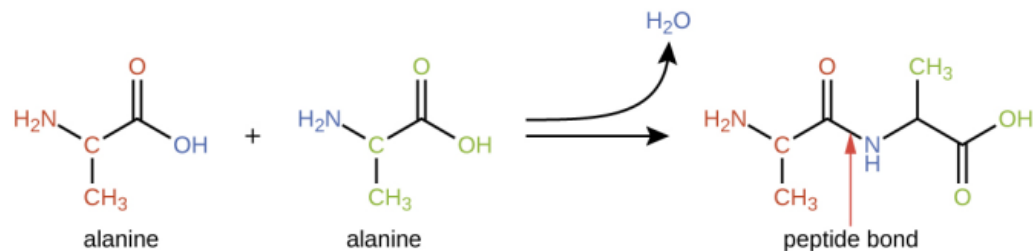


Figure 7.18 Peptide bond formation is a dehydration synthesis reaction. The carboxyl group of the first amino acid (alanine) is linked to the amino group of the incoming second amino acid (alanine). In the process, a molecule of water is released.

✓

✓

- ✓ Molecules formed by chemically linking relatively modest numbers of amino acids (approximately 50 or fewer) are called peptides, and prefixes are often used to specify these numbers: dipeptides (two amino acids), tripeptides (three amino acids), and so forth.
- ✓ More generally, the approximate number of amino acids is designated: **oligopeptides** are formed by joining up to approximately 20 amino acids, whereas **polypeptides** are synthesized from up to approximately 50 amino acids.
- ✓ When the number of amino acids linked together becomes very large, or when multiple polypeptides are used as building subunits, the macromolecules that result are called **proteins**. The continuously variable length (the number of monomers) of these biopolymers, along with the variety of possible R groups on each amino acid, allows for a nearly unlimited diversity in the types of proteins that may be formed.
- ✓ Protein structure is categorized in terms of four levels: primary, secondary, tertiary, and quaternary. The primary structure is simply the sequence of amino acids that make up the polypeptide chain.
- ✓ The chain of amino acids that defines a protein's primary structure is not rigid, but instead is flexible because of the nature of the bonds that hold the amino acids together. When the chain is sufficiently long, hydrogen bonding may occur between amine and carbonyl functional groups within the peptide backbone (excluding the R side group), resulting in localized folding of the polypeptide chain into helices and sheets. These shapes constitute a protein's secondary structure.
- ✓ The most common secondary structures are the α -helix and β -pleated sheet. In the α -helix structure, the helix is held by hydrogen bonds between the oxygen atom in a carbonyl group of one amino acid and the hydrogen atom of the amino group that is just four amino acid

units farther along the chain. In the β -pleated sheet, the pleats are formed by similar hydrogen bonds between continuous sequences of carbonyl and amino groups that are further separated on the backbone of the polypeptide chain

- ✓ The next level of protein organization is the tertiary structure, which is the large-scale three-dimensional shape of a single polypeptide chain. Tertiary structure is determined by interactions between amino acid residues that are far apart in the chain. A variety of interactions give rise to protein tertiary structure, such as disulfide bridges, which are bonds between the sulfhydryl ($-SH$) functional groups on amino acid side groups; hydrogen bonds; ionic bonds; and hydrophobic interactions between nonpolar side chains. All these interactions, weak and strong, combine to determine the final three-dimensional shape of the protein and its function.
- ✓ The process by which a polypeptide chain assumes a large-scale, three-dimensional shape is called protein folding. Folded proteins that are fully functional in their normal biological role are said to possess a native structure.
- ✓ A protein loses its three-dimensional shape, it may no longer be functional. These unfolded proteins are denatured. Denaturation implies the loss of the secondary structure and tertiary structure (and, if present, the quaternary structure) without the loss of the primary structure.
- ✓ Some proteins are assemblies of several separate polypeptides, also known as protein subunits. These proteins function adequately only when all subunits are present and appropriately configured. The interactions that hold these subunits together constitute the quaternary structure of the protein. The overall quaternary structure is stabilized by relatively weak interactions. Hemoglobin, for example, has a quaternary structure of four

globular protein subunits: two α and two β polypeptides, each one containing an iron-based heme

- ✓ Another important class of proteins is the conjugated proteins that have a nonprotein portion. If the conjugated protein has a carbohydrate attached, it is called a glycoprotein. If it has a lipid attached, it is called a lipoprotein. These proteins are important components of membranes. Figure 7.23 summarizes the four levels of protein structure.

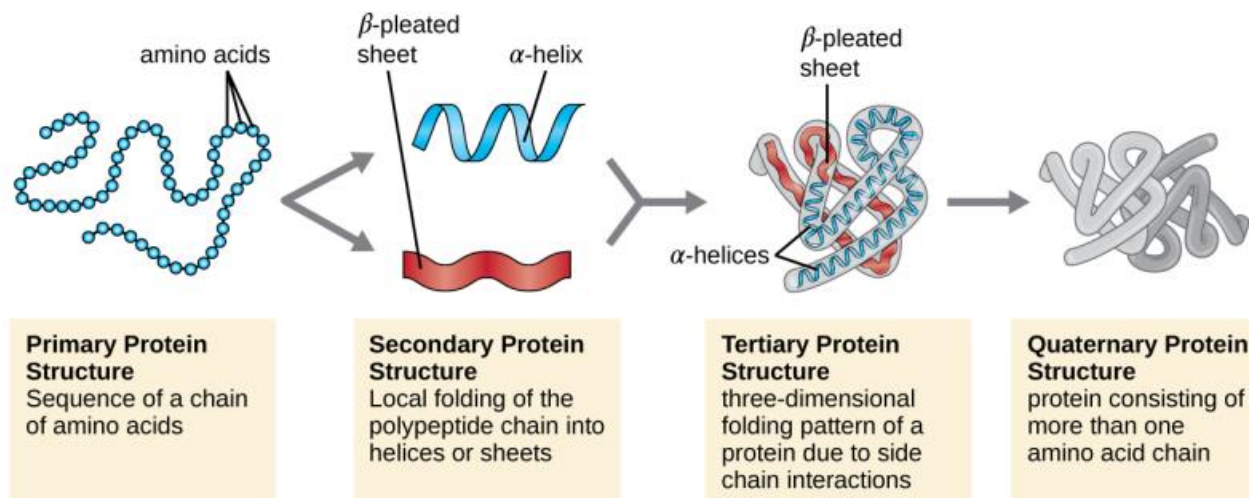


Figure 7.23 Protein structure has four levels of organization. (credit: modification of work by National Human Genome Research Institute)

NUCLEIC ACIDS

nucleic acids are composed of monomers, called **nucleotides**, which are polymerized to form large strands. Each nucleic acid strand contains certain nucleotides that appear in a certain order within the strand, called its **base sequence**. The base sequence of **deoxyribonucleic acid (DNA)** is responsible for carrying and retaining the hereditary information in a cell

DNA Nucleotides

The building blocks of nucleic acids are nucleotides. Nucleotides that compose DNA are called **deoxyribonucleotides**. The three components of a deoxyribonucleotide are a five-carbon sugar called deoxyribose, a phosphate group, and a **nitrogenous base**, a nitrogen-containing ring structure that is responsible for complementary base pairing between nucleic acid strands (**Figure 10.11**). The carbon atoms of the five-carbon deoxyribose are numbered 1', 2', 3', 4', and 5' (1' is read as "one prime"). A nucleoside comprises the five-carbon sugar and nitrogenous base.

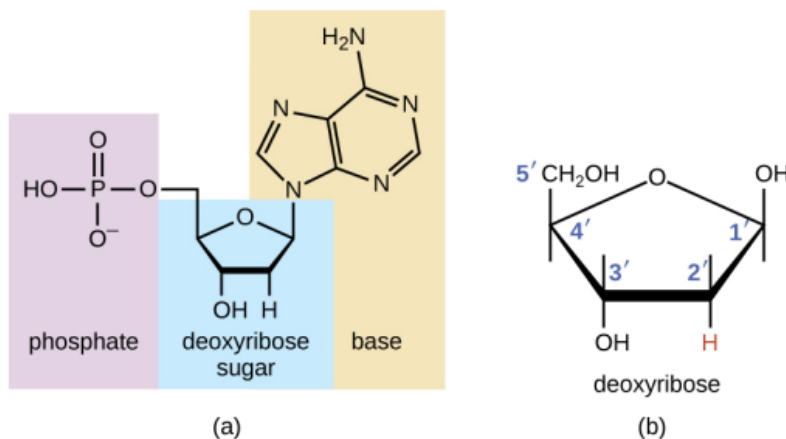


Figure 10.11 (a) Each deoxyribonucleotide is made up of a sugar called deoxyribose, a phosphate group, and a nitrogenous base—in this case, adenine. (b) The five carbons within deoxyribose are designated as 1', 2', 3', 4', and 5'.

The deoxyribonucleotide is named according to the nitrogenous bases (**Figure 10.12**). The nitrogenous bases adenine (A) and guanine (G) are the purines; they have a double-ring structure with a six-carbon ring fused to a five-carbon ring. The pyrimidines, cytosine (C) and thymine (T), are smaller nitrogenous bases that have only a six-carbon ring structure.

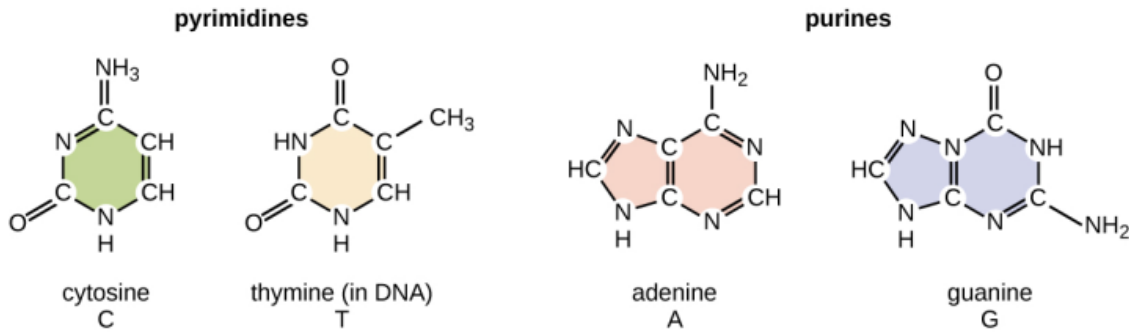


Figure 10.12 Nitrogenous bases within DNA are categorized into the two-ring purines adenine and guanine and the single-ring pyrimidines cytosine and thymine. Thymine is unique to DNA.

- ✓ Individual nucleoside triphosphates combine with each other by covalent bonds known as 5'-3' phosphodiester bonds, or linkages whereby the phosphate group attached to the 5' carbon of the sugar of one nucleotide bonds to the hydroxyl group of the 3' carbon of the sugar of the next nucleotide.
- ✓ Phosphodiester bonding between nucleotides forms the sugar-phosphate backbone, the alternating sugar-phosphate structure composing the framework of a nucleic acid strand (Figure 10.13).
- ✓ During the polymerization process, deoxynucleotide triphosphates (dNTP) are used. To construct the sugar-phosphate backbone, the two terminal phosphates are released from the dNTP as a pyrophosphate. The resulting strand of nucleic acid has a free phosphate group at the 5' carbon end and a free hydroxyl group at the 3' carbon end.
- ✓ The two unused phosphate groups from the nucleotide triphosphate are released as pyrophosphate during phosphodiester bond formation. Pyrophosphate is subsequently hydrolyzed, releasing the energy used to drive nucleotide polymerization.

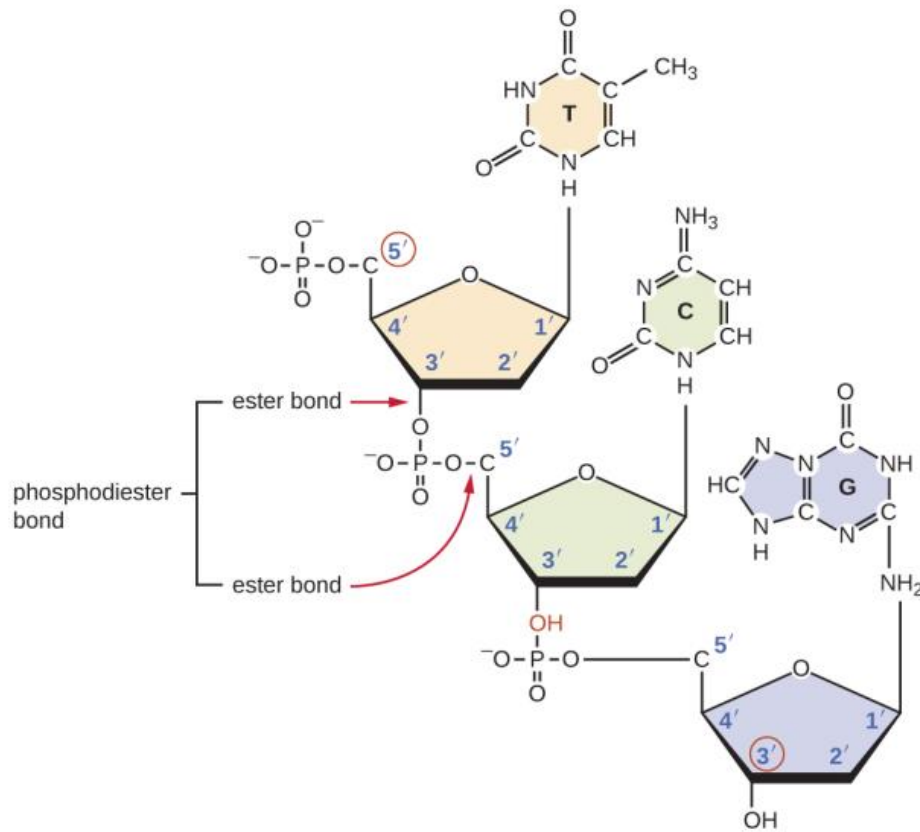


Figure 10.13 Phosphodiester bonds form between the phosphate group attached to the 5' carbon of one nucleotide and the hydroxyl group of the 3' carbon in the next nucleotide, bringing about polymerization of nucleotides into nucleic acid strands. Note the 5' and 3' ends of this nucleic acid strand.

DNA Structure

- ✓ Watson and Crick proposed that DNA is made up of two strands that are twisted around each other to form a righthanded helix. The two DNA strands are **antiparallel**, such that the 3' end of one strand faces the 5' end of the other (**Figure 10.16**). The 3' end of each strand has a free hydroxyl group, while the 5' end of each strand has a free phosphate group. The sugar and phosphate of the polymerized nucleotides form the backbone of the structure, whereas the nitrogenous bases are stacked inside. These nitrogenous bases on the interior of the molecule interact with each other, base pairing.
- ✓ Analysis of the diffraction patterns of DNA has determined that there are approximately 10 bases per turn in DNA. The asymmetrical spacing of the sugar-phosphate backbones generates major grooves (where the backbone is far apart) and minor grooves (where the backbone is close together) (**Figure 10.16**). These grooves are locations where proteins

can bind to DNA. The binding of these proteins can alter the structure of DNA, regulate replication, or regulate transcription of DNA into RNA.

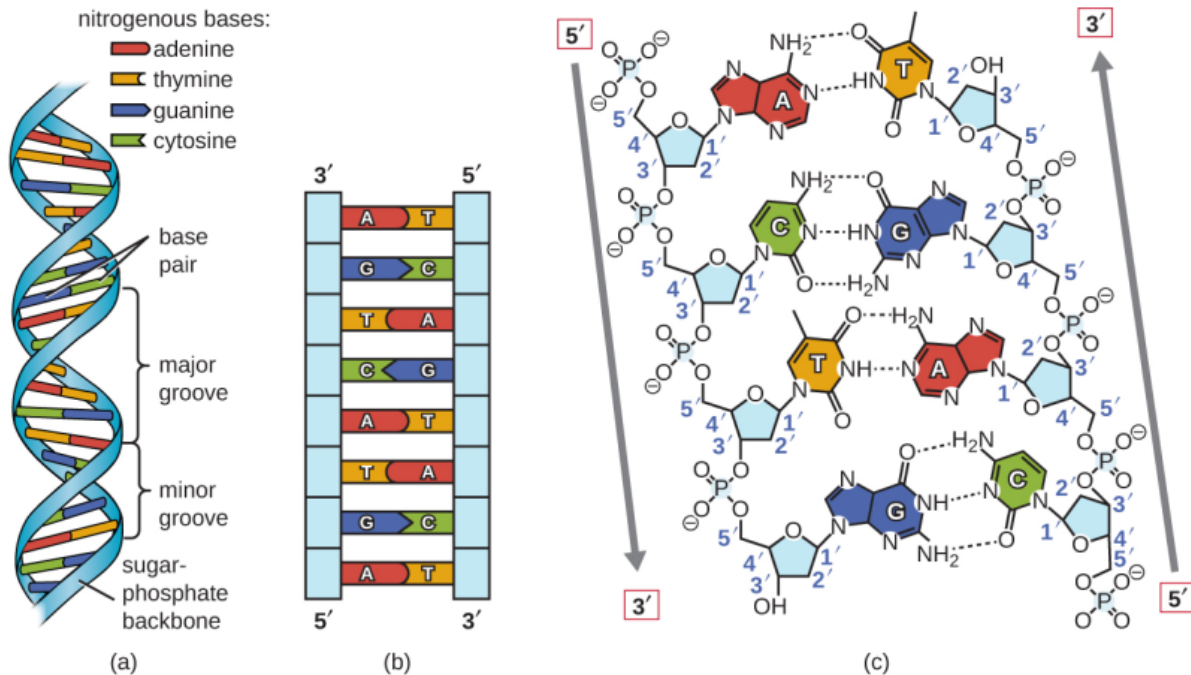


Figure 10.16 Watson and Crick proposed the double helix model for DNA. (a) The sugar-phosphate backbones are on the outside of the double helix and purines and pyrimidines form the "rungs" of the DNA helix ladder. (b) The two DNA strands are antiparallel to each other. (c) The direction of each strand is identified by numbering the carbons (1 through 5) in each sugar molecule. The 5' end is the one where carbon #5 is not bound to another nucleotide; the 3' end is the one where carbon #3 is not bound to another nucleotide.

Base pairing takes place between a purine and pyrimidine. In DNA, adenine (A) and thymine (T) are **complementary base pairs**, and cytosine (C) and guanine (G) are also complementary base pairs. The base pairs are stabilized by hydrogen bonds; adenine and thymine form two hydrogen bonds between them, whereas cytosine and guanine form three hydrogen bonds between them.

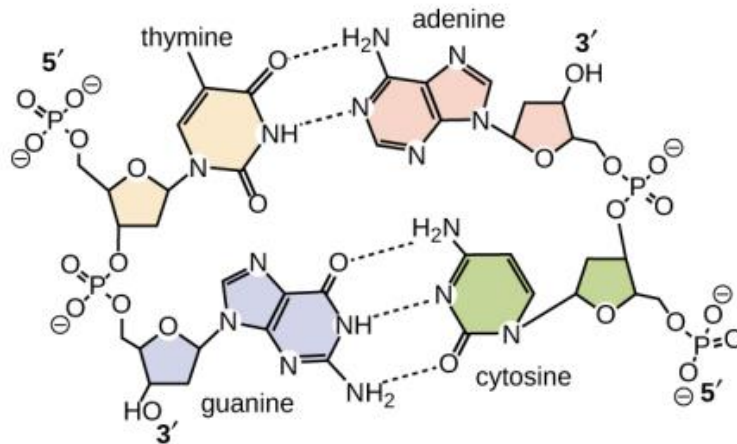


Figure 10.17 Hydrogen bonds form between complementary nitrogenous bases on the interior of DNA.

In the laboratory, exposing the two DNA strands of the double helix to high temperatures or to certain chemicals can break the hydrogen bonds between complementary bases, thus separating the strands into two separate single strands of DNA (single-stranded DNA [ssDNA]). This process is called DNA denaturation and is analogous to protein denaturation, as described in **Proteins**. The ssDNA strands can also be put back together as double-stranded DNA (dsDNA), through reannealing or renaturing by cooling or removing the chemical denaturants, allowing these [hydrogen bonds to reform](#).

RNA Structure

Structurally speaking, **ribonucleic acid (RNA)**, is quite similar to DNA. However, whereas DNA molecules are typically long and double stranded, RNA molecules are much shorter and are typically single stranded. RNA molecules perform a variety of roles in the cell but are mainly involved in the process of protein synthesis (translation) and its regulation.

RNA is typically single stranded and is made of **ribonucleotides** that are linked by phosphodiester bonds. A ribonucleotide in the RNA chain contains ribose (the pentose sugar), one of the four nitrogenous bases (A, U, G, and C), and a phosphate group. The subtle structural difference between the sugars gives DNA added stability, making DNA more suitable for storage of genetic

information, whereas the relative instability of RNA makes it more suitable for its more short-term functions. The RNA-specific pyrimidine **uracil** forms a complementary base pair with adenine and is used instead of the thymine used in DNA. Even though RNA is single stranded, most types of RNA molecules show extensive intramolecular base pairing between complementary sequences within the RNA strand, creating a predictable three-dimensional structure essential for their function (**Figure 10.20** and **Figure 10.21**).

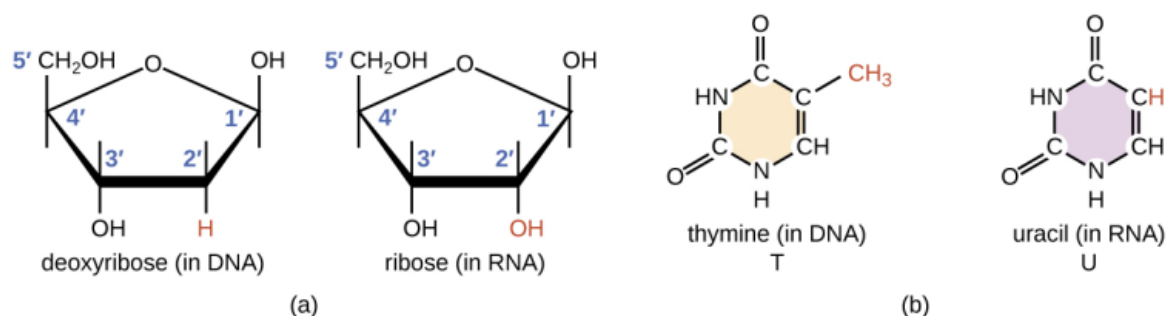


Figure 10.20 (a) Ribonucleotides contain the pentose sugar ribose instead of the deoxyribose found in deoxyribonucleotides. (b) RNA contains the pyrimidine uracil in place of thymine found in DNA.

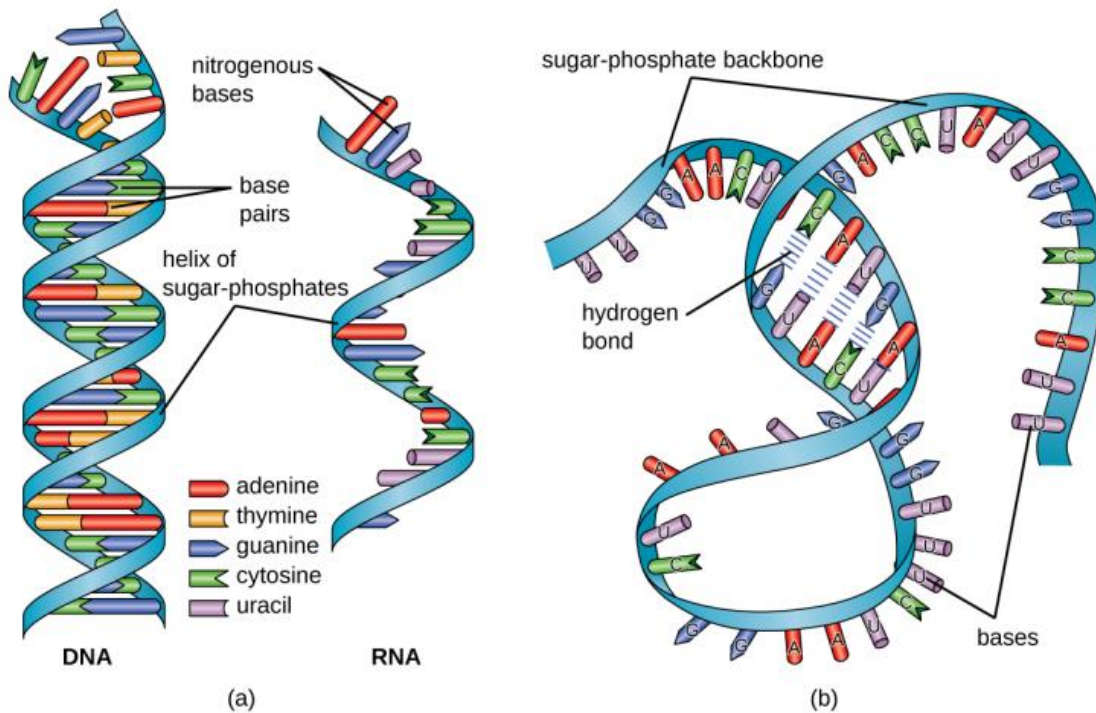


Figure 10.21 (a) DNA is typically double stranded, whereas RNA is typically single stranded. (b) Although it is single stranded, RNA can fold upon itself, with the folds stabilized by short areas of complementary base pairing within the molecule, forming a three-dimensional structure.

Functions of RNA in Protein Synthesis

- ✓ Cells access the information stored in DNA by creating RNA to direct the synthesis of proteins through the process of translation.
- ✓ Proteins within a cell have many functions, including building cellular structures and serving as enzyme catalysts for cellular chemical reactions that give cells their specific characteristics.
- ✓ The three main types of RNA directly involved in protein synthesis are **messenger RNA (mRNA)**, **ribosomal RNA (rRNA)**, and **transfer RNA (tRNA)**.
- ✓ In 1961, French scientists François Jacob and Jacques Monod hypothesized the existence of an intermediary between DNA and its protein products, which they called messenger RNA.[16] Evidence supporting their hypothesis was gathered soon afterwards showing that information from DNA is transmitted to the ribosome for protein synthesis using mRNA.
- ✓ If DNA serves as the complete library of cellular information, mRNA serves as a photocopy of specific information needed at a particular point in time that serves as the instructions to make a protein.
- ✓ The mRNA carries the message from the DNA, which controls all of the cellular activities

in a cell. If a cell requires a certain protein to be synthesized, the gene for this product is

“turned on” and the mRNA is synthesized through the process of transcription. The mRNA

then interacts with ribosomes and **other** cellular machinery (**Figure 10.22**) to direct the synthesis of the protein it encodes during the process of translation (see **Protein Synthesis**). mRNA is relatively unstable and short-lived in the cell, especially in prokaryotic cells, ensuring that proteins are only made when needed. rRNA and tRNA are stable types of RNA. In prokaryotes and eukaryotes, tRNA and rRNA are encoded in the DNA, then copied into long RNA molecules that are cut to release smaller fragments containing the individual mature RNA species.

- ✓ Ribosomes are composed of rRNA and protein. As its name suggests, rRNA is a major constituent of ribosomes, composing up to about 60% of the ribosome by mass and providing the location where the mRNA binds. The rRNA ensures the proper alignment of the mRNA, tRNA, and the ribosomes; the rRNA of the ribosome also has an enzymatic activity (peptidyl transferase) and catalyzes the formation of the peptide bonds between two aligned amino acids during protein synthesis
- ✓ Transfer RNA is the third main type of RNA and one of the smallest, usually only 70–90 nucleotides long. It carries the correct amino acid to the site of protein synthesis in the ribosome. It is the base pairing between the tRNA and mRNA that allows for the correct amino acid to be inserted in the polypeptide chain being synthesized

Structure and Function of RNA

	mRNA	rRNA	tRNA
Structure	Short, unstable, single-stranded RNA corresponding to a gene encoded within DNA	Longer, stable RNA molecules composing 60% of ribosome's mass	Short (70-90 nucleotides), stable RNA with extensive intramolecular base pairing; contains an amino acid binding site and an mRNA binding site
Function	Serves as intermediary between DNA and protein; used by ribosome to direct synthesis of protein it encodes	Ensures the proper alignment of mRNA, tRNA, and ribosome during protein synthesis; catalyzes peptide bond formation between amino acids	Carries the correct amino acid to the site of protein synthesis in the ribosome

RNA as Hereditary Information

Although RNA does not serve as the hereditary information in most cells, RNA does hold this function for many viruses that do not contain DNA. Thus, RNA clearly does have the additional capacity to serve as genetic information

Molecular architecture of microbial cells: chemical composition and function of cellular structures and organelles.